

Analog, safety and security

Every technique must discharge a named verification claim.

Bound the analog
claim



Classify safety
evidence



State the adversary

Learning outcomes

Demonstrate these decisions on a real plan row.

- Define an analog model's validity envelope.
- Distinguish unobserved from structurally irrelevant faults.
- State a threat model and its observation boundary.
- Preserve domain-specific qualifiers at sign-off.

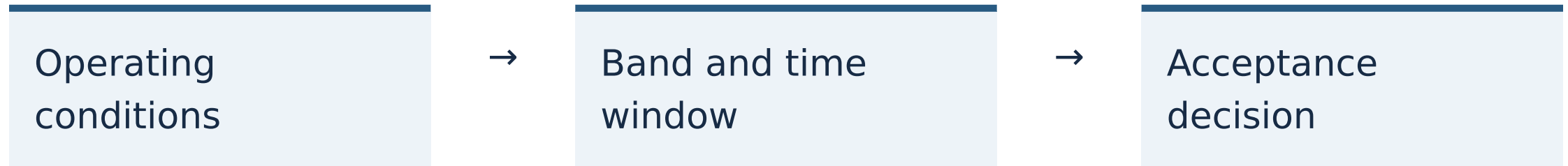
Route through the lesson

Four blocks, each ending with a decision.

- Bound the analog claim
- Classify safety evidence
- State the adversary
- Maintain specialized arguments

An analog oracle needs an envelope

Acceptance depends on tolerance, time and operating conditions.



A fast real-number model omits physics

Its value comes from a declared use boundary.

- Use RNM for represented digital–analog interactions.
- Do not infer unmodeled loading or continuous effects.

Correlation is a maintained obligation

A model comparison needs criteria set before observing the result.

- Compare at named ports and conditions.
- Record tolerances, validity and recheck triggers.

Discussion: the model passed training

Name the evidence that would settle the decision.

- An RNM campaign validates the training algorithm.
- The next claim concerns physical channel variation.
- Can the original verdict discharge it?

Safety adds fault-response obligations

Correct intended behavior is only part of the argument.

- Separate systematic design defects from random hardware faults.
- Connect fault effects to detection and safe response.

Not observed is not a safety classification

A workload can miss a fault effect.

- Distinguish structural irrelevance from lack of excitation.
- Retain the argument behind any exclusion.

Detection has a deadline and a healthy case

A mechanism must react in time without inventing faults.

- Check fault detection and response timing.
- Check acceptable behavior without a fault.

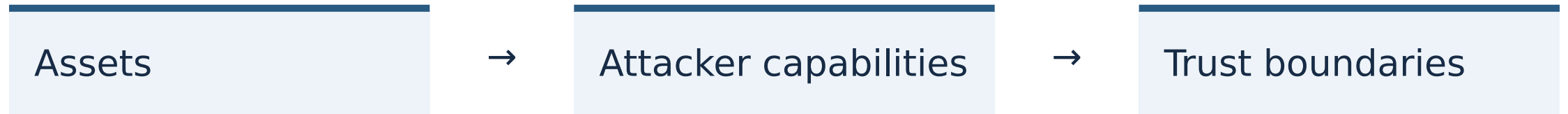
Discussion: unexcited means safe?

Name the evidence that would settle the decision.

- A campaign never activates a modeled fault.
- The row is proposed for removal as harmless.
- What stronger argument would be needed?

Security begins with a threat model

Assets, attacker capability and trust boundaries define the claim.



Functional correctness can coexist with leakage

A correct output does not settle every observation.

- Consider information flow and explicit exceptions.
- Treat release conditions as part of the policy.

A model cannot expose a mechanism it omits

Digital noninterference does not settle physical side channels.

- Choose observations that represent the attack.
- Allocate physical evidence for physical leakage.

Discussion: no path, no leak?

Name the evidence that would settle the decision.

- A structural report shows no direct secret-to-output connection.
- The threat includes timing observations.
- Which property remains unestablished?

Qualification concerns a defined use

Tool identity alone does not describe reliance.

- Record version, configuration and intended use.
- Identify the independent checks supporting reliance.

A shared source can share an omission

Independent evidence must challenge the original interpretation.

- Review model omissions against the claim.
- Revisit assumptions when intended use changes.

Close a specialized row with its qualifiers

Preserve the conditions under which the result is valid.

- Name the model, conditions and observations.
- Keep exclusions and residual claims visible.

Discussion: one green status for every domain

Name the evidence that would settle the decision.

- Analog, safety and security reports are all green.
- Their excluded behaviors are absent from the dashboard.
- What must a sign-off reviewer recover?

Exercise: two unobserved faults

Separate universal irrelevance from limited observation.

- One fault is outside every observation cone.
- Another is never excited by the workload.
- Decide which may be removed and how to handle the other.

What to change on Monday

Make one existing verification decision reviewable.

- Add a validity trigger to a model record.
- Audit an unobserved-fault exclusion.
- Name a security observation missing from the current model.

Scope and next reading

Use the handout for the detailed artifacts and sources.

- Read the Analog, safety and security handout and book Chapters 21–23.
- Normative compliance and full analog or security tool flows are outside this lesson.